APPLICATION NOTE · STRUCTURAL & COMPUTATIONAL BIOINFORMATICS · PREPRINT

charmmgui2cp2k: automated generation of CP2K QM/MM input from CHARMM-GUI/AMBER biomolecular setups

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Open source · MIT | v0.1.0 | Python 3.10–3.12 | CP2K ≥ 7.1

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ABSTRACT

Setting up a biomolecular quantum-mechanics/molecular-mechanics (QM/MM) simulation in CP2K from an existing AMBER / CHARMM-GUI system is a manual, error-prone expert task: the QM region must be enumerated, every boundary link atom defined, boundary charges redistributed, and the AMBER topology pre-edited. We present **charmmgui2cp2k**, a terminal-native wizard (a plain CLI and a full-screen Textual TUI sharing one validated core) that automates this end to end and emits immediately runnable CP2K QM/MM input, embedding the scientific safeguards a careful setup requires — topology validation, hydrogen link atoms with forbidden-cut rejection, charge-conserving boundary schemes, spin-state risk analysis, and CP2K version/data-file capability gating. To our knowledge it is the first automated CHARMM-GUI/AMBER → CP2K QM/MM input generator. We validate correctness by NVE energy conservation, real-CP2K parser acceptance, and an independent ParmEd cross-check, and show the generated boundary is energy-conserving.

Availability and implementation: open source under the MIT license; source, tests, and the reproducible validation harness at the repository (archived on Zenodo at release). **Contact:** [email].

1 Statement of need

CP2K is a widely used engine for biomolecular QM/MM, able to read AMBER `prmtop` and CHARMM PSF topologies directly, but it ships no setup GUI or wizard: users hand-author the `&QMMM` region (`QM_KIND` / `MM_INDEX`), every `&LINK` atom, the boundary charge handling, and Lennard-Jones/charge pre-edits of the topology. Mistakes in any of these silently produce physically wrong simulations.

No existing tool closes this gap end to end (Table 1). CHARMM-GUI's QM/MM Interfacer is web-based, semiempirical-only, and targets the CHARMM/AMBER engines, not CP2K; ASH is a scripting library rather than a guided wizard; MiMiCPy targets CPMD+GROMACS. **charmmgui2cp2k** fills this gap and is delivered as an offline, terminal-native wizard aimed at enzyme active sites, flavoenzymes, metalloproteins, and other redox-cofactor systems — where boundary and charge correctness matter most.

Table 1. Landscape of QM/MM setup tools and the unmet gap.

Tool	Form	Engine target	Closes CHARMM-GUI/AMBER→CP2K gap?
CHARMM-GUI QM/MM Interfacer	web GUI	CHARMM/AMBER, semiempirical	No — no CP2K, no DFT
CP2K (native)	input language	CP2K	No — no setup wizard
ASH	Python library	many (CP2K as a backend)	No — library, no CHARMM-GUI ingest
MiMiCPy	wizard	CPMD + GROMACS	No — different stack
gmx2qmmm	script	GROMACS↔Gaussian/ORCA/…	No — no CP2K
QwikMD	VMD GUI	NAMD	No — no CP2K
charmmgui2cp2k	**CLI + TUI wizard**	**CP2K**	**Yes**

2 Implementation

The pipeline is a single Python module exposing three frontends over one shared scientific core (Figure 1): a plain CLI wizard, a Textual TUI workbench (Figure 2; eight phases:

System → QM region → Boundary → Method → Electronic → Workflow → Preview → Generate), and a non-interactive batch mode. The core performs:

- **Topology validation** — AMBER `POINTERS`/array cross-checks, CHAMBER rejection, mixed-SCEE detection.
- **QM/MM boundary** — automatic detection of bonds crossing the boundary, hydrogen link-atom capping (IMOMM), classification and rejection of chemically impossible cuts, M1/M2 neighbour enrichment, GEEP embedding radii, and generation-time link-geometry sanity checks.
- **Charge conservation** — residual-charge redistribution across split-residue MM atoms with PRMTOP round-trip verification, plus a combined cross-channel audit (FIST residual + ADD_MM_CHARGE embedding).
- **Electronic structure** — GTH-valence electron counting, multiplicity-parity enforcement, and a multi-detector spin-risk taxonomy (transition metals, redox cofactors, π-stacking geometry, unresolved elements) that refuses to auto-assign spin when the choice is unreliable.
- **CP2K capability gating** — keyword/basis/dispersion emission gated against the detected CP2K version *and* against the presence of the corresponding data files (GTH potentials, ADMM basis, dftd3.dat) in the install.
- **Provenance** — every generated input embeds a deterministic parameter header; a full decision log, boundary-charge audit, and electronic-state report are written alongside.

A `--strict` mode turns unresolved scientific concerns into a non-zero exit for reproducible pipelines.

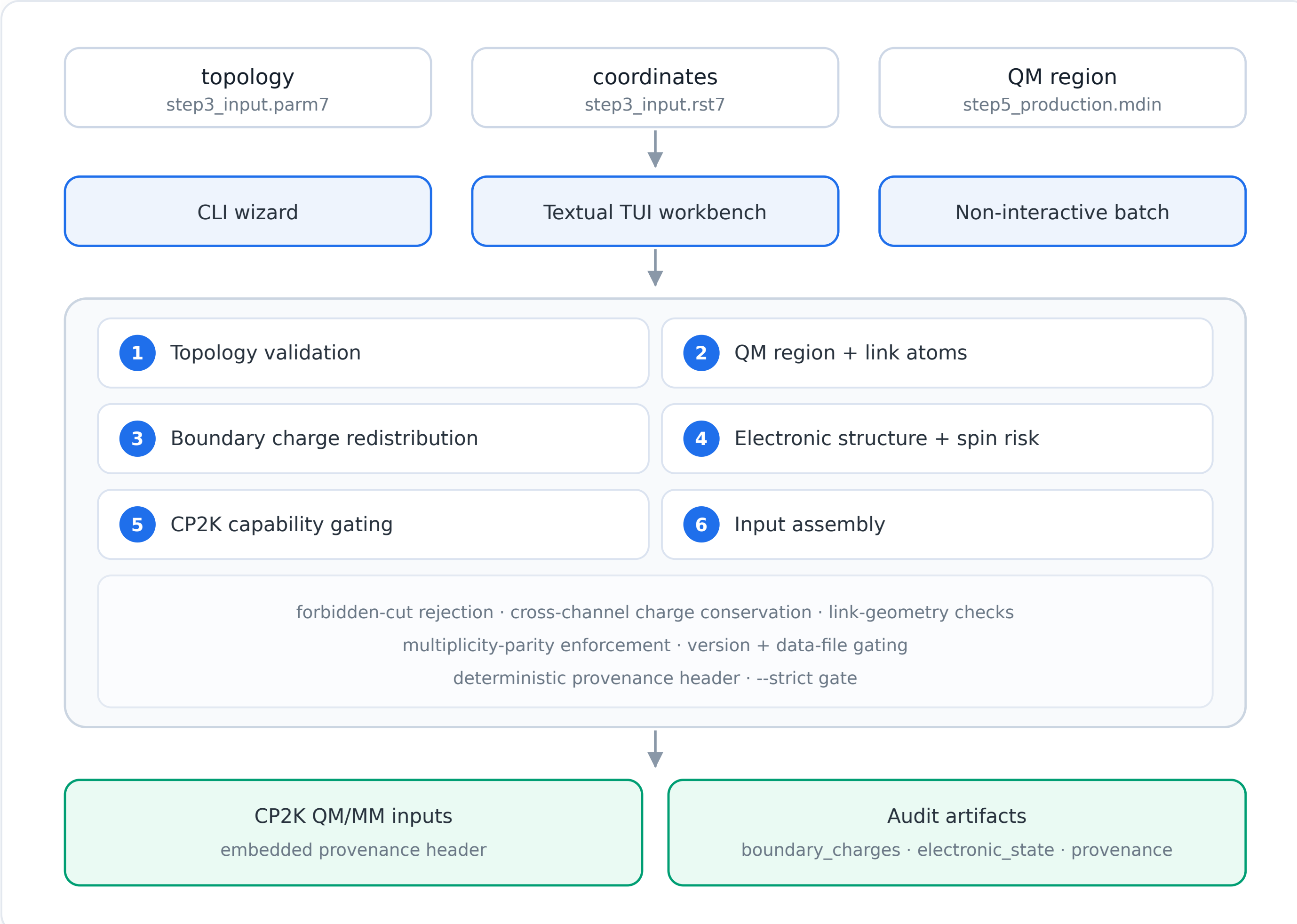
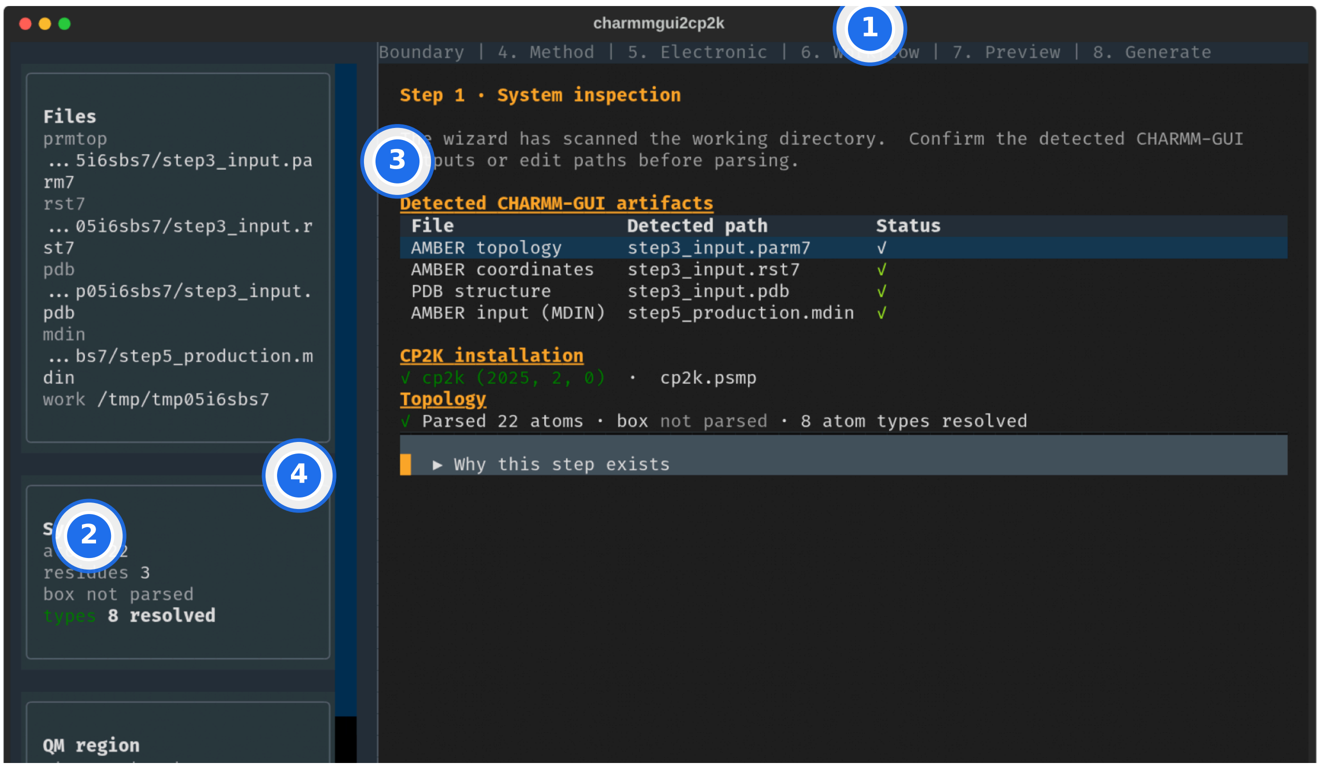


Figure 1. Architecture and data flow. Three frontends share one validated scientific core that ingests CHARMM-GUI/AMBER inputs, applies six sequential stages under cross-cutting safeguards, and emits runnable CP2K QM/MM input plus machine- and human-readable audit artifacts.



- Phase rail — current and remaining steps
- Sticky system summary (atoms, residues, QM region)
- Auto-detected inputs, each with its provenance
- Progressive disclosure — expert detail on demand

Figure 2. The Textual TUI workbench (rendered live, headlessly) on the System phase. Callouts: **(1)** the phase rail showing current and remaining steps; **(2)** the sticky system summary; **(3)** auto-detected inputs, each with its provenance; **(4)** progressive disclosure of expert detail. Keyboard-first navigation makes the next safe action obvious without a manual.

3 Validation

All validation is reproducible from the `validation/` harness on a committed alanine-dipeptide demo system (build script included); production-scale agreement is the flavoenzyme case study (§3.4).

3.1 Real-CP2K acceptance

Every generated staged input passes a real CP2K binary's parser-only `--input-check` (CP2K 2025.2); see `tests/regression/test_cp2k_input_check.py`.

3.2 NVE energy conservation

A self-contained NVE QM/MM trajectory of the generated setup (PBE, 50 fs) gives a conserved-quantity range of 0.104 kcal/mol and a drift of **2.6×10^{-4} kT/dof/ps** with no systematic trend (Figure 3a), demonstrating the link atoms, deleted boundary terms, and boundary charges are mutually consistent. A longer trajectory with tighter SCF tolerance reduces this toward the literature reference (10^{-5}–10^{-6} kT/dof/ps; Götz *et al.*, 2014 [6]).

3.3 Boundary handling and reproducibility

Single-point energies under the NONE / Z1 / CHARGE_SHIFT boundary schemes differ measurably: NONE (the full M1 frontier charge retained in the embedding) lies 2.61 kcal/mol from CHARGE_SHIFT, while Z1 and CHARGE_SHIFT (both remove M1 from the embedding) agree to 0.26 kcal/mol (Figure 3b). This confirms the boundary charge treatment is physically active and that the redistribution schemes control the frontier over-polarization NONE exhibits. Repeated runs are bitwise reproducible ($|\Delta E| = 5\times10^{-15}$ Ha). Element identities and charges agree with an independent ParmEd parse (`tests/unit/test_parmed_crosscheck.py`).

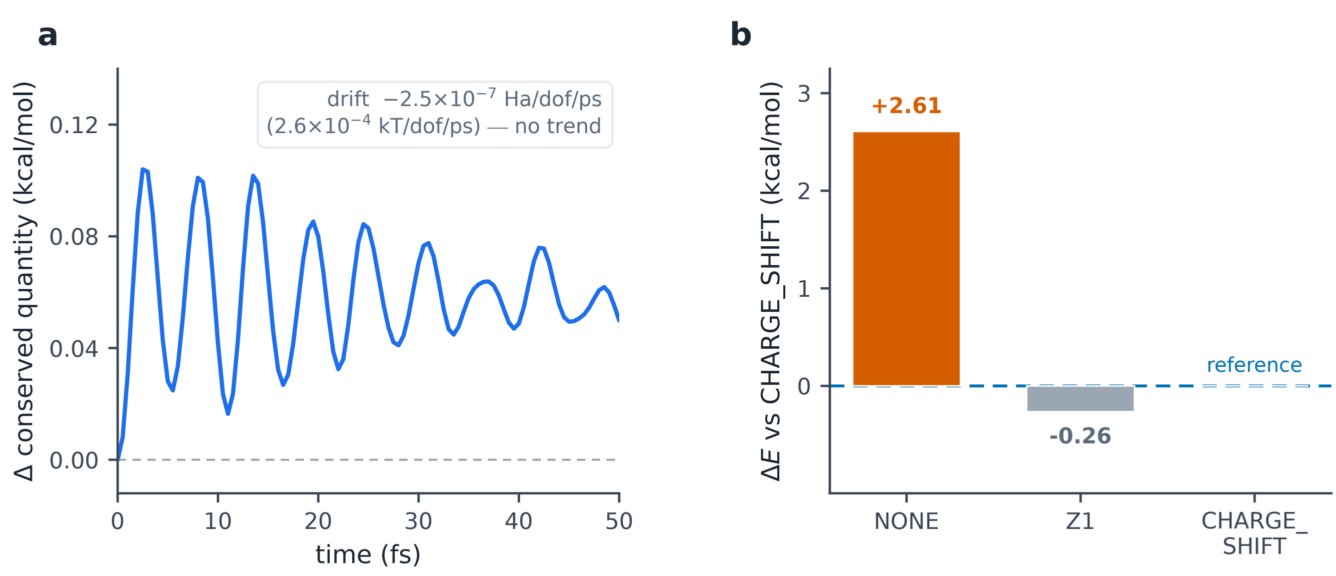


Figure 3. Validation on the alanine-dipeptide demo (PBE; CP2K 2025.2). **(a)** NVE energy conservation — the conserved quantity oscillates within ~0.1 kcal/mol and the transient decays; bounded, with no systematic drift (2.6×10^{-4} kT/dof/ps). **(b)** Boundary-scheme over-polarization — single-point QM/MM energy relative to CHARGE_SHIFT: NONE (full M1 charge in the embedding) is an outlier by +2.61 kcal/mol, while Z1 and CHARGE_SHIFT (both remove M1 from the embedding) agree to within 0.26 kcal/mol.

3.4 Flavoenzyme case study DATA PENDING

Headline validation on a flavoenzyme active site (L-amino-acid oxidase, LAAO; a redox-active FAD cofactor), where the QM region spans the flavin and substrate and both boundary/charge handling and spin-state assignment are most scrutinized: single-point and short-MD agreement against a reference, plus a boundary over-polarization probe. The harness runs unchanged via `--dir`; results will be inserted here.

3.5 Benchmark suite DATA PENDING

Agreement with the BioExcel GROMACS+CP2K QM/MM benchmark systems (MQAE, ClC, CBD_PHY, GFP) to show tool-generated CP2K input matches standalone references.

4 Usability and reproducibility (FAIR)

The primary workflow is designed to need no external manual: defaults trace to source evidence, expert controls use progressive disclosure, validation failures state the corrective action, and the generation phase shows progress and final artifacts. Each generated input embeds its full parameter set as a comment header (FAIR R1.2 [15]); a one-command `--demo` reproduces a complete run from bundled data. The software ships an OSI license, a `CITATION.cff`, and continuous-integration tests; releases will be archived on Zenodo for a versioned DOI.

5 Availability

Source code, tests, and the validation harness are available under the MIT license at [repository URL — on publication], archived on Zenodo for a versioned DOI. Tested on Linux with Python 3.10–3.12 and CP2K ≥ 7.1.

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Editorial note (preprint). Author names/affiliations and the repository DOI are placeholders pending submission; §3.4–3.5 await the production metalloprotein system and the external benchmark suite. Reference details should be verified against the publishers' records before submission.

 Reproducible: `figures/make_figures.py` **Figures embedded**

charmmgui2cp2k — automated CHARMM-GUI/AMBER→CP2K QM/MM input generation. Self-contained manuscript compiled 2026-06-24. Figures 3–4 are generated directly from the committed validation output; Figure 2 is a live headless render of the TUI.